How temperatures near the equinox synchronize spring biological events and the climate change

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Many biological processes, including plant leafout and flowering, occur once cumulative temperatures reach a threshold (the thermal-sum model). In this way, temperatures are thought to coordinate the timing of biological events. But growing evidence suggests that as climates warm, both the advancement of spring has slowed (declining sensitivity) and the variance in the timing of spring events has increased (declining synchrony)—raising questions about the resilience of temperature-based coordination to anthropogenic climate change. To answer these questions, researchers have complicated the thermal-sum model, introducing additional factors and mechanisms (e.g., chilling and photoperiod). We consider whether such complexity is necessary. Using results from the theory of stopped random walks, we show that sensitivity and synchrony are exactly as predicted by the thermal-sum model. In particular, the theory predicts that as temperatures increase and springtime events shift from the equinox toward the winter solstice, the events themselves become less coordinated and more variable. We verify these predictions using experimental and real-world data, including 10,000 observations of common lilacs (United States, 1956-2025). We find that the apparently complex current shifts in temperature sensitivity and variance of phenology with climate change can be predicted by the thermal-sum model. Thus, any efforts to add complexity thus need to explain more than this simple model.

Introduction

Many biological events are triggered not by temperature on a single day, but by temperature accumulated over many days (Larcher, 1980). For example, the buds of flowering plants first bloom in the spring once cumulative temperatures exceed a plant-specific threshold (Quetelet, 1849). This mechanism is thought to synchronize a wide variety of seasonal events within an ecosystem—for instance, ensuring that pollinators emerge when flowers are in bloom. It also supports an array of forecasting tools in which scientists track cumulative temperatures or 'growing degree days' to predict harvest dates, manage pest emergence, and assess how changing temperature patterns reshape ecological communities (Schwartz, 2024).

Shifts in the timing of biological events have been among the clearest fingerprints of anthropogenic climate change across both natural and managed systems

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(Pörtner et al., 2022). But these shifts have proven difficult to fully predict or explain. One puzzle is that, while the timing of many biological events has moved earlier (i.e., advanced) as climates warm, the rate of advancement has slowed in recent years (Fu et al., 2015). We refer to this pattern in which events appear less sensitive to additional warming as 'declining sensitivity.'

A second puzzle is that the variance in the timing of biological events has in some cases declined and in others increased (Vitasse et al., 2018; Stemkovski et al., 2023). Higher variance has also manifested as previously unknown or rare events becoming more common (Chuine et al., 2025). We refer to this pattern in which events appear less coordinated as 'declining synchrony.'

This paper asks two questions. First, are declining sensitivity and synchrony consistent with the thermal-sum model in which an event occurs once cumulative temperature reaches a threshold? Second, if the thermal-sum model is broadly correct, what does it imply about how these patterns will evolve as climates continue to warm?

To answer these questions, we first show that the biological thermal-sum model is an example of a stopped random walk, and we propose a corresponding statistical framework. We then apply the framework to both observational and experimental data.

We find that declining sensitivity and declining synchrony are in fact consistent with the thermal-sum model. This finding challenges a growing literature in which those patterns are taken as evidence that the thermal-sum model is incorrect and additional factors or mechanisms are needed (Fu et al., 2015; Wolkovich et al., 2021; Gao et al., 2024).

More importantly, the statistical framework reveals the key mechanism that determines whether synchrony declines: the temperature regime under which accumulation occurs. For example, when temperatures increase day over day, as they often do near the equinox, springtime event tend to be tightly synchronized. But when warming shifts those events closer to the solstice, where temperatures are comparatively stationary, event times become more variable.

The thermal-sum model predicts that continued warming will result in the widespread transition from a 'coordinated spring' to a 'scattered spring.' We find that under the CMIP6 scenario of high warming (SSP5-8.5), the synchrony of lilac bloom dates, a common indicator of spring, will decline rapidly in Southern United States by the year 2100.

Results & Discussion

The thermal-sum model states that a biological event occurs when cumulative temperature reach a threshold. We first show this model is an example of a stopped random walk, and we approximate the distribution of the event times using statistical arguments commonly applied in such cases (Siegmund, 1967; Woodroofe, 1982; Gut, 2009). This theoretical result provides a statistical framework that we then apply to both observational and experimental data. The Appendix provides additional details on the derivation, simulations, and data construction.

To derive the theoretical result, consider the timing of a specific biological event, such as the bloom date of a bud. Let X_i denote the effective temperature experienced by that bud on day i. We model X_i under the two temperature regimes illustrated in Figure 1.

The first regime is the stationary temperature regime in which the average daily temperature is approximately constant at some level $\alpha > 0$. This regime is most often observed in the winter months following the solstice, and we refer to temperatures in this regime as 'winter temperatures.' The second regime is the increasing temperature regime in which the average daily temperature rises at a constant rate $\beta > 0$. This regime is most often observed in the spring months closer to the equinox, and we refer to it as 'spring warming.' In both regimes, effective temperature is modeled as average temperature plus noise ϵ_i. Thus, $X_i = \alpha + \beta i + \epsilon_i$, where $\beta = 0$ in the winter-temperature regime and $\beta > 0$ in the spring-warming regime.

A more elaborate model could represent seasonal temperatures with a sinusoidal trend, for example, but for our purposes the two-regime specification with a fixed transition captures the necessary features of the problem. In Figure 1, the effective temperature is assumed to be above 0°C since biological limits typically prevent the accumulation of low ambient temperatures, although this may also be adjusted.

Noise in this formulation can represent the influence of various biological factors, depending on the scale of analysis and on our evolving understanding of what triggers springtime events. At the bud level, it represents idiosyncratic variation specific to that bud, which could arise from microclimate differences (Schwartz et al., 2014), bud color variation (Peaucelle et al., 2022), or internal biological differences that affect when the bud begins accumulating temperatures (van der Schoot et al., 2014; Pan et al., 2023). We assume the ϵ_i are independent with mean 0 and

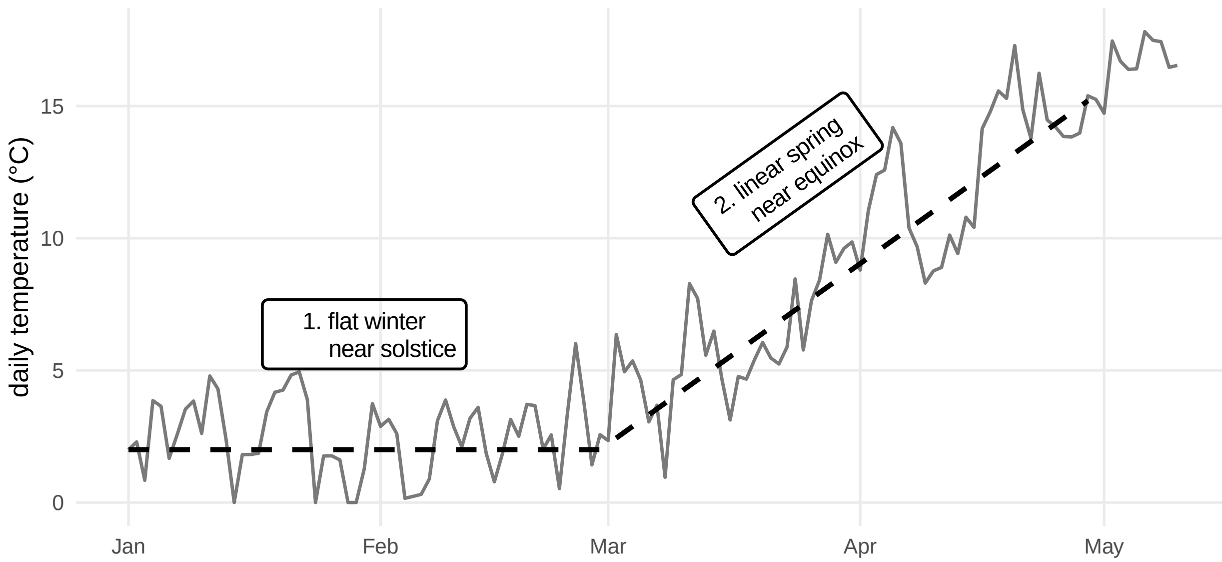

Figure 1: We model the daily temperature (solid line) as trend (dashed line) plus noise. The timing of springtime events in temperate climates depends on whether temperature is accumulated during one of two regimes, 1. the stationary temperature regime (relatively flat near the winter solstice), which we refer to as 'winter temperatures,' or 2. the increasing temperature regime (relatively linear near the spring equinox), which we refer to as 'spring warming.'

variance σ^2, although this assumption can be relaxed.

The bloom date $\nu(\tau)$ is the day n on which cumulative temperature $Z_n = \sum_{i=1}^{n} X_i$ first reaches a plant-specific threshold $\tau > 0$, that is, $\nu(\tau) = \min\{m : Z_m > \tau\}$. This formulation is standard in growing degree day models and, more generally, Z_n is the "forcing" component of many process-based models of plant phenology (Schwartz, 2024). Standard results for stopped random walks yield asymptotically normal approximations for the bloom date $\nu(\tau)$ when the threshold τ is sufficiently large that accumulation occurs over many days. See Appendix for details.

These approximations show that the mean and variance of bloom dates depend on the underlying temperature regime (i.e., values of α and β) during which the plant accumulates temperature. In the winter temperatures regime, average temperatures are constant during accumulation ($\beta = 0$), and cumulative temperature grows at an approximately linear rate. It follows that the expected bloom date advances with climate change in step with increasing winter temperatures, $\mathbb{E}[\nu(\tau)] \approx \tau/\alpha$, as is often described in the literature (Vitasse et al., 2022). In the spring warming regime, average temperatures increase during accumulation ($\beta > 0$), and cumulative temperatures grow quadratically. In this regime, the expected bloom date squared advances in step with spring warming, $\mathbb{E}[\nu(\tau)] \approx \sqrt{2\tau/\beta}$.

Within either regime, the relationship between bloom dates and temperature is non-linear as observed in recent papers (Fu et al., 2015; Wolkovich et al., 2021). Indeed, both regimes predict declining sensitivity in that the average bloom date advances at a diminishing rate when warming increases as measured by the winter temperatures α or spring warming β (for the winter temperatures regime: $\frac{\partial}{\partial \alpha}\,\mathbb{E}[\nu(\tau)] \approx -\tau/\alpha^2$; for the spring warming regime: $\frac{\partial}{\partial \beta}\,\mathbb{E}[\nu(\tau)] \approx -\frac{1}{2}\sqrt{2\tau/\beta^3}$).

Importantly, the effects of warming within the two regimes make very different predictions regarding the synchrony of events—that is the variance in event timings within the same year and location due to idiosyncratic differences described above such as microclimate. Both predict decreased variance with warming (for the winter temperatures regime: $\mathrm{Var}(\nu(\tau)) \approx \sigma^2\,\tau/\alpha^3$; for the spring warming regime: $\mathrm{Var}(\nu(\tau)) \approx \sigma^2/\sqrt{2\beta^3\tau}$). But the overall size of the variance depends on whether the threshold τ is in the numerator or the denominator.

Note that if accumulation occurs over many days, then τ must be large relative to α and β. Thus when the threshold τ is in the denominator as in the spring warming regime, the variance is small—reflecting the fact that noise is quickly overwhelmed by the quadratic increase in cumulative temperatures that occurs during spring warming. This is how temperatures near the equinox synchronize plants (and buds within plants) with the same threshold—making idiosyncratic differences such as microclimate negligible. In contrast, when the threshold τ is in the numerator as in the winter temperatures regime, the variance is large—reflecting the fact that noise remains relatively influential in that idiosyncratic differences can translate into larger differences in threshold-crossing times.

The fact that the variance is predicted to be small in the spring warming regime and larger in the winter temperatures regime suggests that the observed variance will increase when anthropogenic warming shifts accumulation from the spring warming regime to the winter temperatures regime. The shift in regimes is observable as a 'scattered spring' in which, as springtime events advance, they become less coordinated.

Not all warming scatters spring, however. Warming can lead to a more synchronized or more scattered spring depending on the specific values of α, β, and τ. In some locations, climate change may manifest primarily as higher winter temperatures (an increase in α) with relatively little change in the rate at which temperatures rise during spring (a modest change in spring warming β). In others, winter temperatures may change less while spring transitions steepen (a larger increase in spring warming β).

The theoretical results presented here offer a simple way to translate spatially varying climate change into spatially varying predictions of both the advancement of spring and its variability. To demonstrate the framework, we analyze bloom dates from the lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015), collected across the continental United States from 1956–2014. We limit our analysis to the purple common lilac (*Syringa vulgaris*) and the phenophase full bloom, and we update the dataset using USA National Phenology Network records from the same sites and species until 2025 (Switzer et al., 2025).

For each record, we estimate winter temperatures α and spring warming β. We then estimate the conditional mean and log standard deviation functions of the bloom date given α and β using a generalized additive model (Hastie and Tibshirani, 1986; Wood, 2017). The two functions are displayed in Figure 2. Select cross sections of these functions are displayed in Figure 3. Note that the bloom date is measured in days since January 1 of the observation year.

These two figures show that increasing either winter temperatures α, spring warming β, or both shift flowering times exactly as predicted by the theory. Increasing α or β is associated with earlier mean bloom dates, which generally but not always advance at a decreasing rate. Increasing α and holding β fixed returns greater variance (higher log standard deviation), while increasing β while holding α fixed returns lower variance (lower log standard deviation). This is the 'scattered spring' predicted from the theory: When β is fixed, increasing α means more buds bloom in the winter temperatures regime, under which conditions the variance is predicted to be larger. When α is fixed and β is increased, the percentage of buds blooming in either regime does not change. Rather the fixed percentage of buds that bloom under spring warming are subject to increased warming, and under these conditions the variance is predicted to be smaller.

Under the proposed framework, the relationship between biological event times and temperatures are reduced to two parameters, α and β. Projected values of α and β can be translated to projected bloom date distributions as we demonstrate in Figures 4 and 5. We calculate α and β in the years 2050, 2075, and 2100 under temperature projections from NEX-GDDP-CMIP6 scenarios of moderate warming (SSP2-4.5) and high warming (SSP5-8.5) (Thrasher et al., 2022). We then estimate the mean and standard deviation of lilac bloom dates subject to those temperatures using the generalized additive model described above.

Figures 4 and 5 show that under the moderate warming scenario (SSP2-4.5), the mean and standard deviation are relatively unchanged. However, under the high warming scenario (SSP5-8.5), the mean bloom date advances moderately while the standard deviation increases dramatically. In particular, the standard deviation increases from a week to a month in parts of the Southern United States, an indication of the 'scattered spring' that occurs when the temperature regime shifts.

We also apply the framework to experimental data from Charrier et al. (2011), who tested the effect of different constant temperatures on walnut (*Juglans*~sp.) tree budburst after the plants had experienced sufficient chilling to break endodormancy (stems were sampled from 30 trees, chilled at 4°C) . Since temperatures are held constant, this experiment replicates the winter temperatures regime in which the bloom date is approximately normal with mean τ/α and variance $\sigma^2\,\tau/\alpha^3$, which we fit using weighted least squares. The data fit the model well, exhibiting the nonlinear relationship in the mean and variance as predicted by the theory (Figure 6), further supporting the thermal-sum model can explain observed trends, even under the extreme temperature range and controlled chilling

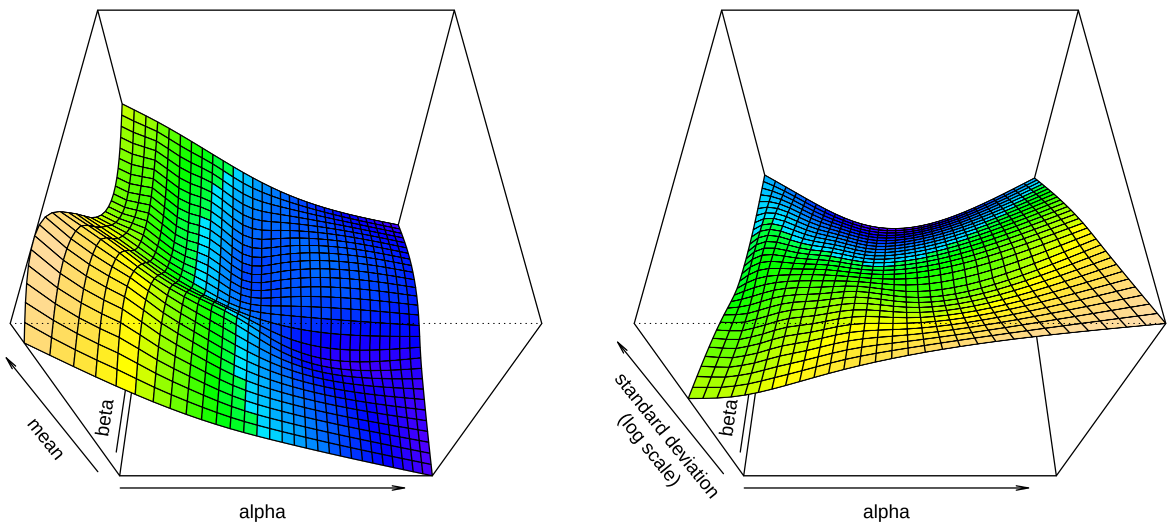

Figure 2: Mean and log standard deviation of lilac bloom dates (n = 10,613 from Rosemartin et al. 2015, updated to 2025 using USA National Phenology Network data) by winter temperatures α and spring warming β as estimated by a generalized additive model. The mean (left) decreases as α and β increase. The standard deviation (right) can increase or decrease depending on the relative values of α and β.

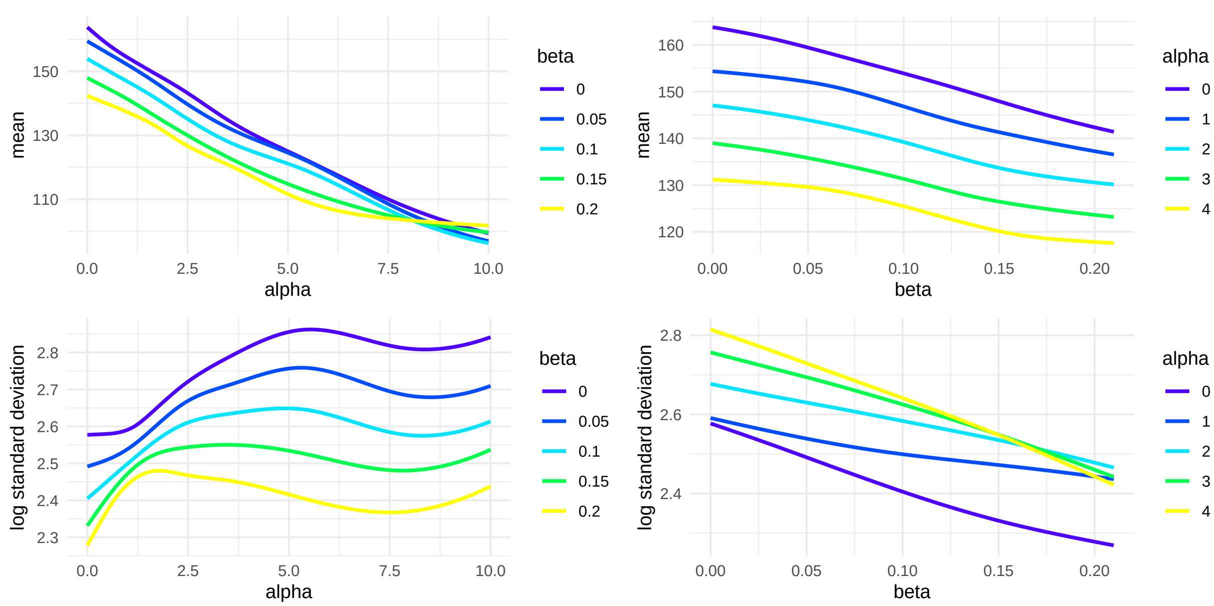

Figure 3: Mean (top) and log standard deviation (bottom) of lilac bloom dates (n = 10,613 from Rosemartin et al. 2015, updated to 2025 using USA NPN records) by winter temperatures α and spring warming β as estimated by a generalized additive model. The top two figures show the mean generally decreases as α increases holding β constant (left) and β increases holding α constant (right). The bottom two figures show the standard deviation can increase or decrease depending on whether α increases with β held constant (left) or β increases with α held constant (right).

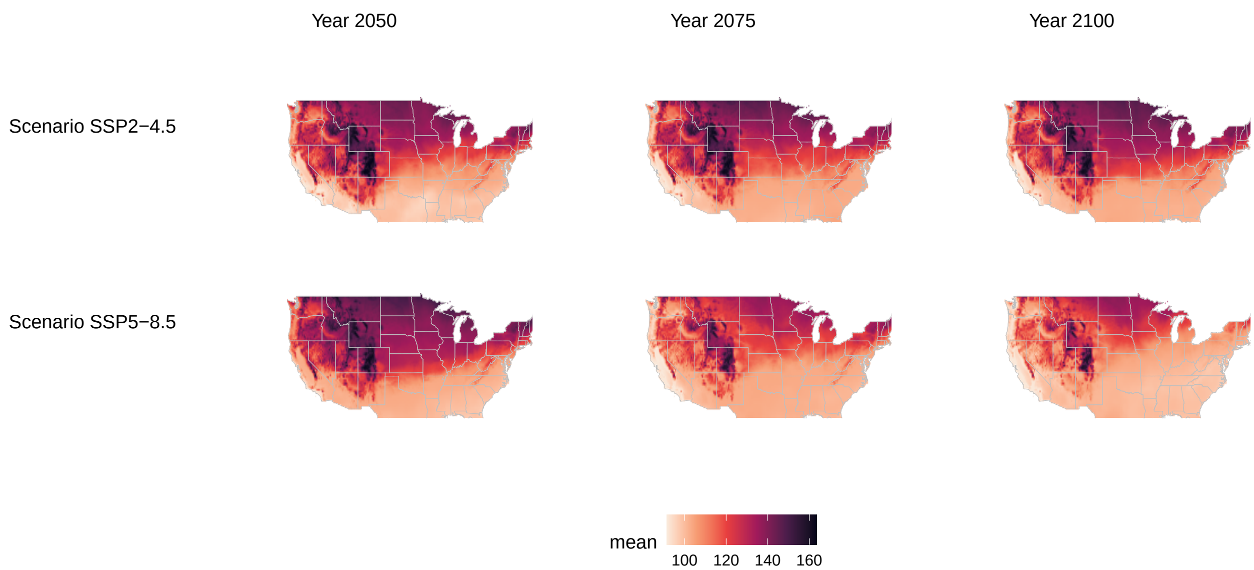

Figure 4: Predicted mean of lilac bloom dates in years 2050, 2075, and 2100 (columns) across the continental United States under temperatures projected under two scenarios (rows) from NEX-GDDP-CMIP6 (Thrasher et al., 2022) in which additional warming is moderate (top) or high (bottom). Lighter areas indicate earlier blooms due to higher temperatures. Under SSP2-4.5, the mean bloom date is largely constant between 2050 and 2100, while under SSP5-8.5, the mean bloom date advances moderately.

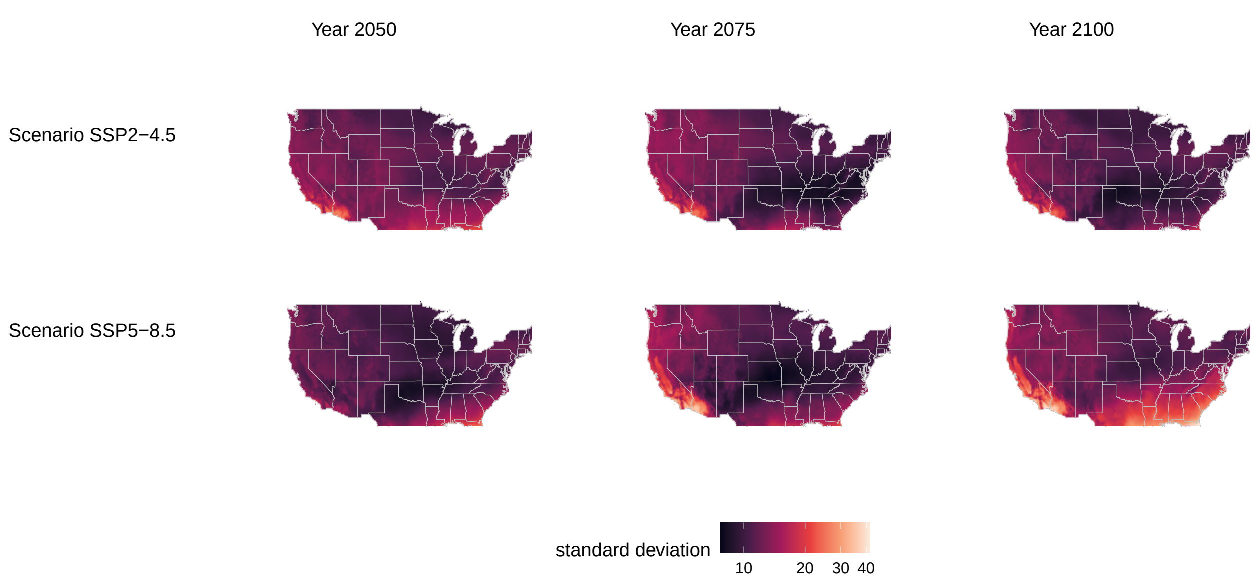

Figure 5: Predicted standard deviation of lilac bloom dates in years 2050, 2075, and 2100 (columns) across the continental United States under temperatures projected under two scenarios (rows) from NEX-GDDP-CMIP6 (Thrasher et al., 2022) in which additional warming is moderate (top) or high (bottom). Light areas indicate more variation (less synchrony) due to higher temperatures. Under SSP2-4.5, the standard deviation is largely constant between 2050 and 2100, while under SSP5-8.5, the standard deviation increases dramatically, particularly in the Southeast and Southwest. These dramatic increase is due to a regime change that we refer to as a 'scattered spring.'

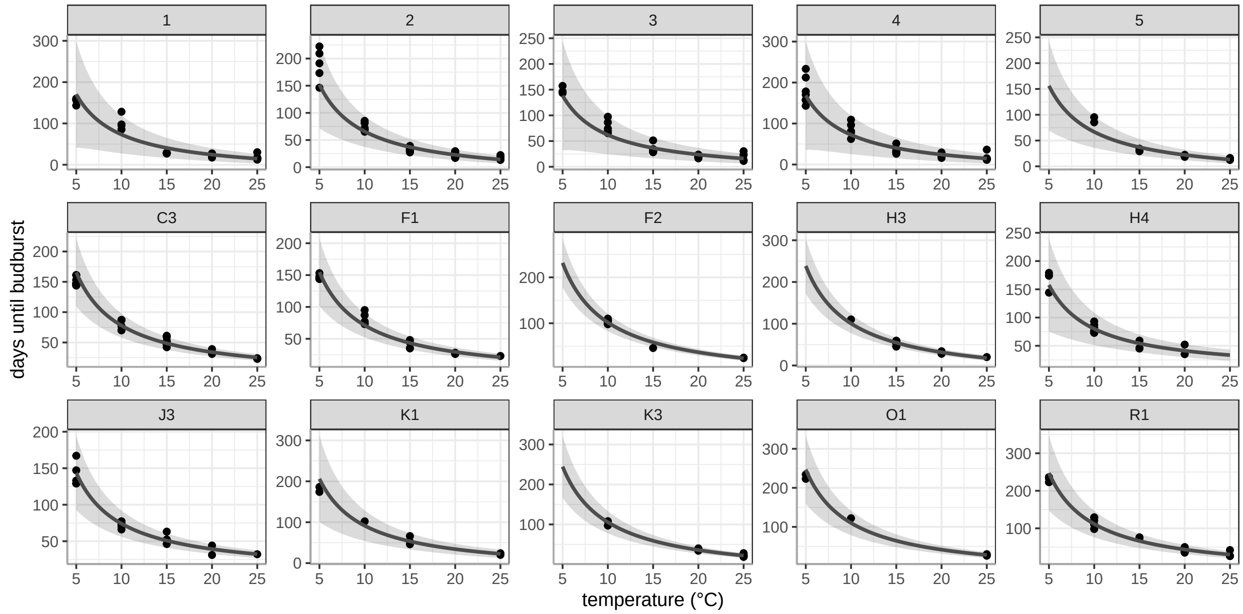

Figure 6: Mean time until budburst as a function of the average daily temperature (alpha) in a controlled experiment conducted by Charrier et al. (2011) in which stems were sampled from 30 walnut trees (*Juglans*~sp.) trees, chilled at 4°C, and then warmed in different temperatures environments (x-axis) with the same photoperiod conditions across temperatures until budburst. We show the observed data (points), mean (solid line), 95% confidence interval (inner grey region, approximately two standard errors), and 95% prediction interval (outer grey region, approximately two standard deviations). The figure shows that the mean and variance are nonlinear functions of temperature even when biological factors such as chilling and photoperiod are held fixed. Both decrease with increasing temperatures as predicted by the thermal-sum model.

and photoperiod conditions in experiments.

We conclude that the thermal-sum model explains observed shifts in the timing of biological events. In particular, it explains observed declines in sensitivity and synchrony. This finding has two important consequences. First, it challenges an extensive body of work that has attempted to revise the thermal-sum model by adding additional factors or mechanisms such as photoperiod or chilling (Fu et al., 2015; Kovaleski, 2021). These factors can of course influence the timing of biological events, but our work suggests they do not themselves explain variation in sensitivity and synchrony since the experimental data we consider exhibits these behaviors even when factors like photoperiod and chilling are held constant.

To understand the role played by these or any additional factors, the basic relationship between event time and temperature must be modeled correctly. Indeed, the literature often assumes a linear relationship between temperature and event time with constant variance, which we have shown is inconsistent with the thermal-sum model. The framework we propose is one way to encode the thermal-sum model, on top of which complexity may be added.

The second consequence is that the thermal-sum model predicts a 'scattered spring' or general desynchronization of springtime events as their timing moves closer to the winter solstice. For example, we have shown that the synchrony of lilac bloom dates, a common indicator of spring, will decline rapidly in Southern United States by the year 2100 under scenario SSP5-8.5, although species with lower τ may decline sooner. Absent some compensating mechanism, the timing of biological events across the ecosystem—from crops to forests, and from plants to insects—will in many cases become more variable, with potential cascading consequences for industry and natural ecosystems.

Methods

Our observational data analysis considers all sites in the historical lilac–honeysuckle phenology network (Rosemartin et al., 2015) that recorded the phenophase full bloom and were within 10 miles of a Global Historical Climatology Network daily (GHCNd) site with sufficiently complete temperature records. We limit our analysis to common lilac (*Syringa vulgaris*). We include all observations made from 1955 to 2025 as recorded by the USA National Phenology Network, retrieved using the `R` package `rnpn`, resulting in $10,613$ observations.

For each observation, we estimate α using the average daily temperature between January and February. We estimate β using the slope of a linear regression model fit by regressing daily temperature on day index over March and April. Temperatures below 0 are set to 0, a common base temperature in growing degree day models.

The parameters α and β are shown for nine sites in Figures 7-9 in Appendix A.3. Figure 7 shows nine randomly selected site-years and has the same interpretation as Figure 1. Site 14726 in New Mexico (Year 1969) is a good example of a bloom in the spring warming regime since relatively little temperature was accumulated during January and February. The average bloom date was May 8th with a standard deviation of 17 days. Site 14276 in New York (Year 1982) is a good example of a bloom between the winter temperatures and spring warming regimes, which had an average bloom date of March 18 with a standard deviation of 27. Figures 8 and 9 show α and β for the nine sites with the longest running records.

We fit a two-parameter Gaussian location–scale model using the `gaulss` function from the `R` package `mgcv` in which we define

$$\mu(\alpha, \beta) = \mathbb{E}[\nu(\tau) \mid \alpha, \beta]$$

$$\sigma(\alpha, \beta) = \mathrm{SD}(\nu(\tau) \mid \alpha, \beta).$$

Both μ and σ are modeled with tensor-product splines with basis dimension k = 10. The fitted functions are shown in Figures 2 and 3. We then calculate $\tilde{\alpha}$ and $\tilde{\beta}$ for the years 2050, 2075, and 2100 using the NEX-GDDP-CMIP6 temperature projections for two scenarios: SSP2-4.5 and SSP5-8.5 (Thrasher et al., 2022). The projected mean $\mu(\tilde{\alpha}, \tilde{\beta})$ and standard deviation $\sigma(\tilde{\alpha}, \tilde{\beta})$ are shown in Figures 4 and 5.

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A. Appendix

A.1 Sketch of Theoretical Results

We follow the notation in Gut (2009). Let $X_i = \alpha + \beta i + \epsilon_i$ denote the temperature on day i, where $\alpha > 0$, $\beta \geq 0$, and the $\{\epsilon_i\}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$ and $\mathrm{Var}(\epsilon_i) = \sigma^2$. Define the deterministic cumulative sum

$$\xi_n = \sum_{i=1}^{n} \mathbb{E}[X_i] = \alpha n + \frac{\beta}{2} n(n+1) = \frac{\beta}{2} n^2 + \beta\gamma n,$$

where $\gamma = \frac{\alpha}{\beta} + \frac{1}{2}$ and $\xi_n = \alpha n$ when $\beta = 0$.

Denote the sum of the stochastic component $S_n = \sum_{i=1}^{n} \epsilon_i$. The object of interest is the first-passage time of $Z_n = S_n + \xi_n$,

$$\nu(\tau) = \min\{n \geq 1 : \; Z_n > \tau\}.$$

In the language of Lai and Siegmund (1977), $\{Z_n\}$ is a perturbed random walk. That is, the random walk $\{S_n\}$ is perturbed by the deterministic term $\{\xi_n\}$. This setup yields two asymptotic regimes, corresponding to the empirical distinction between the winter temperatures regime in which $\beta \approx 0$ and the spring warming regime in which $\beta > 0$.

When $\beta = 0$, we apply the usual asymptotic argument for the hiting time of a random walk with constant positive drift as $\tau \to \infty$,

$$\nu(\tau) \,\dot{\sim}\, \mathrm{Normal}\left(\frac{\tau}{\alpha},\; \frac{\sigma^2\,\tau}{\alpha^3}\right),$$

see Gut (2009).

When $\beta > 0$, the deterministic component $\xi_n = \frac{\beta}{2} n^2 + \beta\gamma n$ grows quadratically so that the threshold τ is reached near the deterministic crossing time $m(\tau)$ satisfying $\xi_{m(\tau)} \approx \tau$. The solution is

$$m(\tau) = \frac{-\beta\gamma + \sqrt{\beta^2\gamma^2 + 2\beta\tau}}{\beta} = \sqrt{\frac{2\tau}{\beta}} - \gamma + o(1).$$

By the functional central limit theorem, the deviation of $\nu(\tau)$ about $m(\tau)$ is approximately normal. Linearization of $Z_{\nu(\tau)} \approx \tau$ around $m(\tau)$ yields

$$\nu(\tau) \,\dot{\sim}\, \mathrm{Normal}\left(m(\tau),\; \frac{\sigma^2\, m(\tau)}{\left(\alpha + \beta\, m(\tau)\right)^2}\right) \;\approx\; \mathrm{Normal}\left(\sqrt{\frac{2t}{\beta}} - \gamma,\; \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right),$$

where we have used the fact that $\alpha + \beta m(\tau) \approx \beta m(\tau) \approx \sqrt{2\beta\tau}$ when τ is large.

It follows that in the spring warming regime, the mean hitting time scales like $\sqrt{\tau}$ (rather than linearly in τ as in the winter temperatures regime) and the variance shrinks at rate $\tau^{-1/2}$ (rather than growing at τ as in the winter temperatures regime).

A.2 Additional Tables

In addition to the model-based approach in the main text, we also examine the data using a binning approach. The results are contained in Tables 1-3.

Table 1 shows data from a controlled experiment on walnut trees (*Juglans*~sp.) conducted by Charrier et al. (2011). The data examines 30 trees comprising 6 genotypes at 2 locations. In November of the study year, stems were sampled from each tree and cut into 7 centimeter segments containing a single bud. All segments were chilled at 4°C and then transferred to constant 'forcing' environments with temperatures set at one of $5, 10, 15, 20,$ or 25°C. For each temperature, the outcome is the response time, the time (in days) until budburst determined from the date at which 50% of buds reached stage 15 of the BBCH scale.

Table 1: Summary data from the Walnut forcing experiment (Charrier et al. 2011).

α	n	mean	sd
5	30	178	27.28
10	30	88	12.44
15	30	39	11.33
20	30	26	7.67
25	30	20	4.45

Table 2: Mean bloom date (day-of-year) from lilac data (Rosemartin et al. 2015, updated to 2025 using USA NPN records) by average Jan-Feb temperature α (rows) and average Mar-Apr temperature increase β (columns).

$\alpha \setminus \beta$	[−0.28, 0.07]	(0.07, 0.11]	(0.11, 0.15]	(0.15, 0.68]
(0, 0.7]	157.70	152.07	147.60	142.20
(0.7, 1.9]	151.12	146.88	141.64	136.15
(1.9, 4.7]	136.30	131.92	127.58	124.94
(4.7, 16.4]	109.56	109.77	107.36	105.71

Table 3: Standard deviation of bloom date (day-of-year) from lilac data (Rosemartin et al. 2015, updated to 2025 using USA NPN records) by winter level α (rows) and spring warming rate β (columns).

$\alpha \setminus \beta$	[−0.28, 0.07]	(0.07, 0.11]	(0.11, 0.15]	(0.15, 0.68]
(0, 0.7]	12.80	11.94	10.90	10.95
(0.7, 1.9]	12.76	11.91	12.67	12.99
(1.9, 4.7]	16.68	14.97	14.45	12.55
(4.7, 16.4]	19.39	17.06	15.31	12.35

This experiment isolates the winter regime because temperature is held constant over time. To show consistency with this regime, Table 1 reports the mean and standard deviation of the response times across the 30 stems at each forcing temperature. Higher forcing temperatures advance budburst at a decreasing rate (the mean falls with α), and the dispersion of budburst timing is smaller at warmer forcing temperatures. Thus, in an experimental setting, the stopped-random-walk approximation captures the basic comparative statics of constant-temperature forcing on both the level and variability of budburst times. This pattern cannot be explained by chinning or photoperiod as these are held constant.

Tables 2-3 show data from the historical lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015), limited to common lilac and updated using data from USA NPN (Switzer et al., 2025) as describd in the main text. We bin sites and years within a grid of baseline temperatures (α, rows) and springtime warming temperatures (β, columns) and calculate the mean and standard deviation of the observed bloom date, as the number of days since January 1. The results are consistent with Figures 2 and 3 in the main text. The mean and variance are nonlinear functions of α and β. In particular, the variance generally increases when α increases (holding β fixed) and decreases when β decreases (holding α fixed).

A.3 Additional Figures

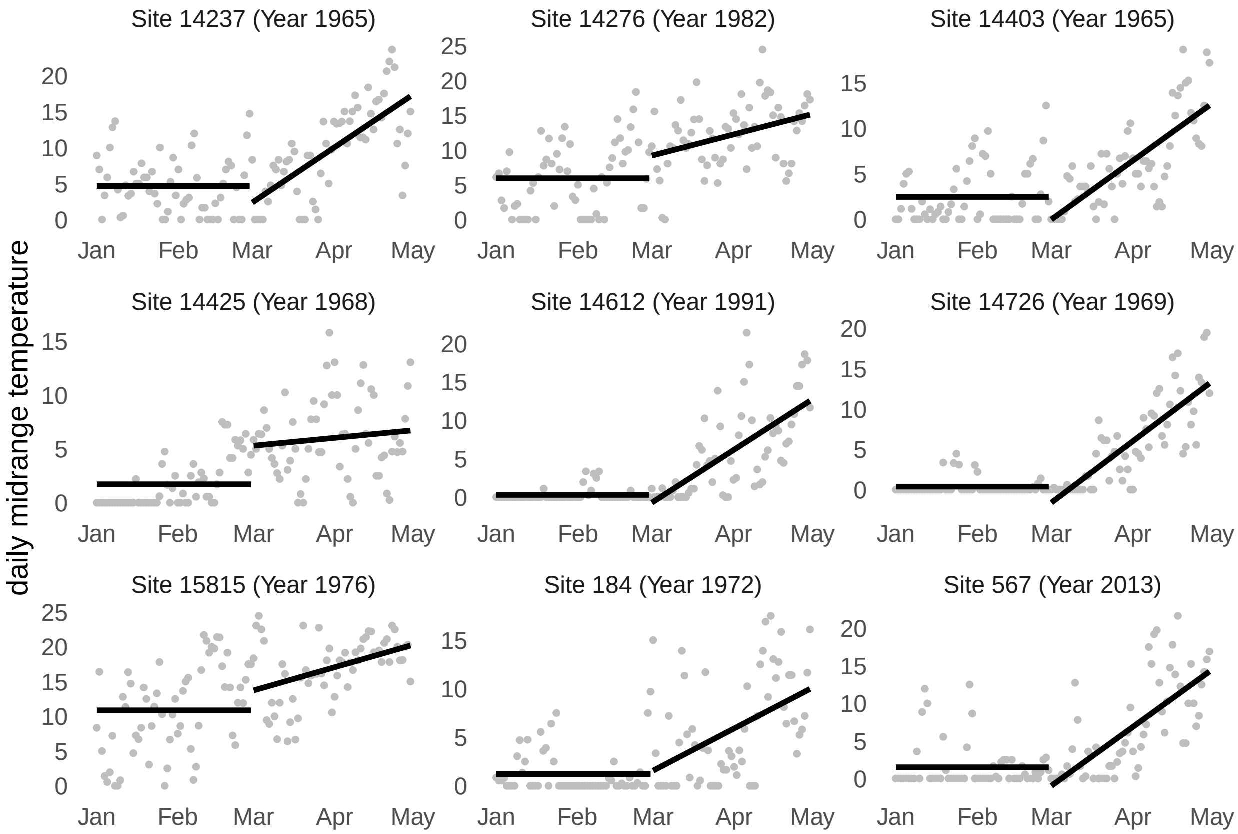

Figure 7: Springtime daily midrange temperatures (points) with α (mean from Jan to Feb) and β (slope from Mar to Apr) for nine randomly selected sites from lilac-honeysuckle dataset (Rosemartin et al. 2015) as measured by GHCND stations nearby. Temperatures below 0 are set to 0.

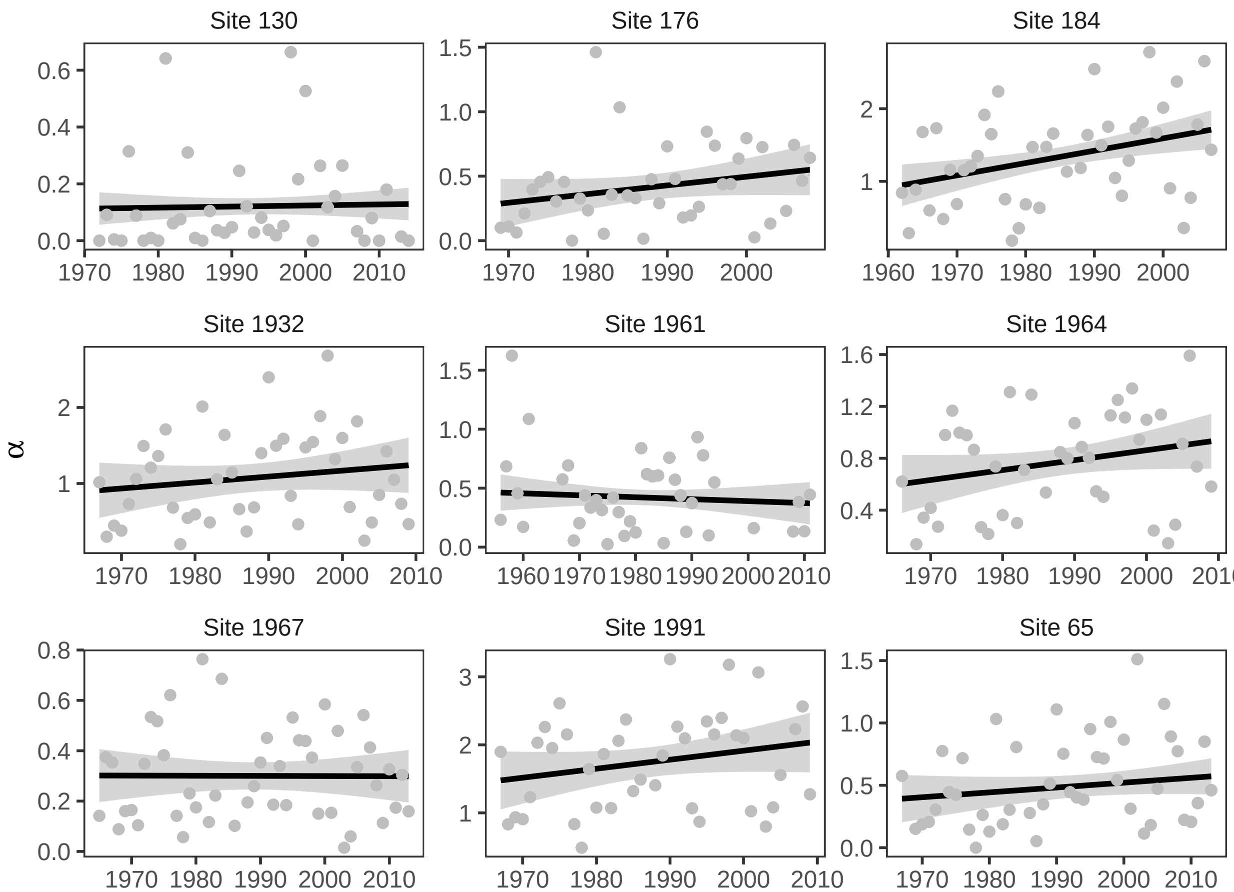

Figure 8: Changes in α for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby. Trend line and confidence interval calculated using linear regression.

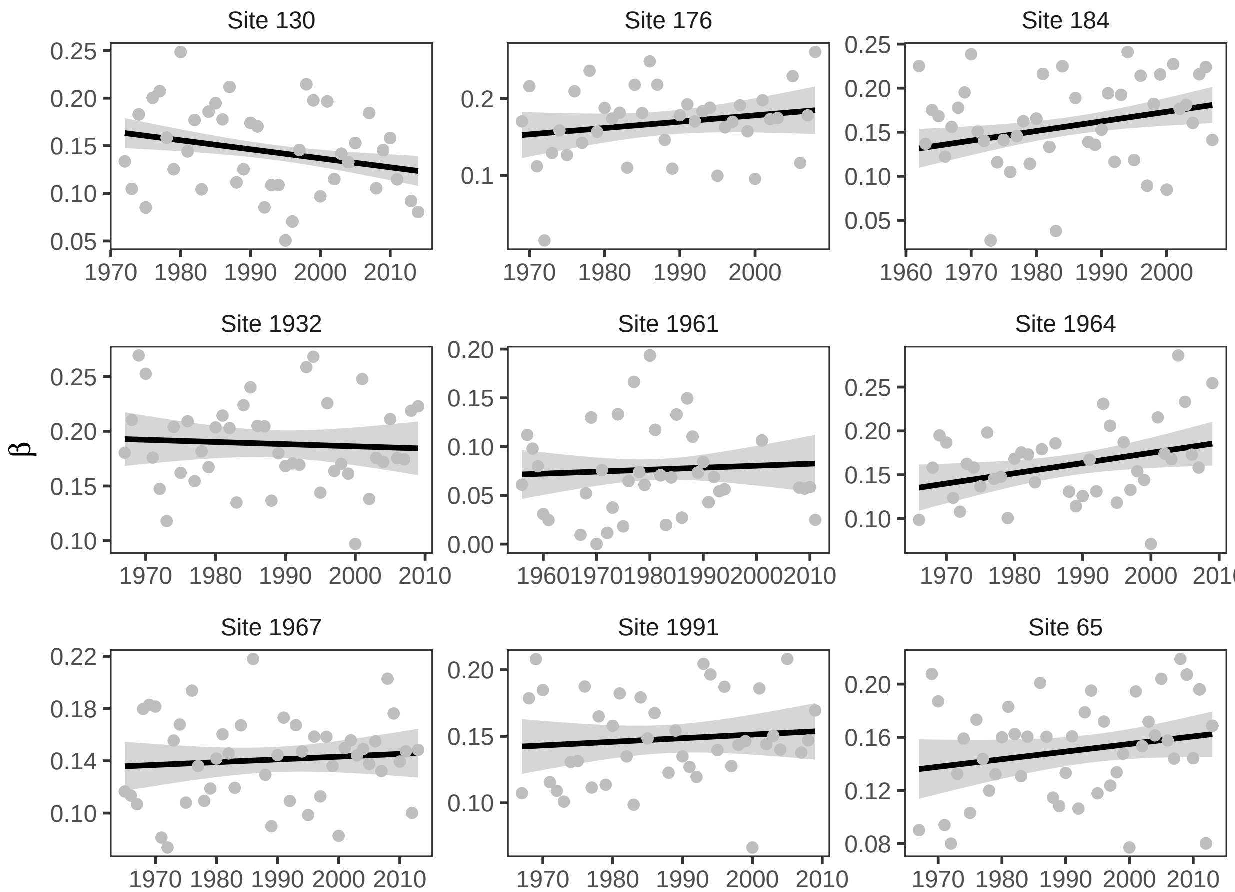

Figure 9: Changes in β for the nine sites from lilac-honeysuckle data (Rosemartin et al. 2015) with longest running records as as measured by GHCND stations nearby. Trend line and confidence interval calculated using linear regression.

A.4 Simulations

We run two simulations to verify the interpretation of our findings. The first checks the asymptotic approximation derived in Section A.1. The second checks the interpretation of the data in Section A.2.

Simulation 1

The first simulation verifies the accuracy of the two asymptotic normal approximations for the hitting time

$$\nu(\tau) = \min\Big\{n \geq 1 : \sum_{i=1}^{n} X_i > \tau\Big\}$$

under the model

$$X_i = \alpha + \beta i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma^2).$$

We set $\sigma = 20$ and generate daily temperatures until the threshold τ is reached, recording the corresponding hitting time $\nu(\tau)$. We repeat this procedure $R = 10{,}000$ times for thresholds $\tau = \{1000, 2000\}$, $\alpha = \{2, 4\}$ and $\beta = \{0, 0.1\}$. When $\beta = 0$, the simulation matches the winter temperatures regime (regime 1), and when $\beta = 0.1$, it matches the spring warming regime (regime 2).

We then compare these simulations with the corresponding theoretical approximation. When $\beta = 0$, we use the winter temperatures regime approximation

$$\nu(\tau) \mathrel{\dot\sim} \text{Normal}\left(\frac{\tau}{\alpha},\ \frac{\sigma^2\tau}{\alpha^3}\right).$$

When $\beta > 0$, we use the spring warming regime approximation

$$\nu(\tau) \mathrel{\dot\sim} \text{Normal}\left(\sqrt{\frac{2\tau}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2},\ \frac{\sigma^2}{\beta^{3/2}\sqrt{2\tau}}\right).$$

For each setting we computed the standardized variable

$$z = \frac{\nu(\tau) - \mu}{\text{sd}}.$$

The simulated distribution of $\nu(\tau)$ is represented in Figure 10 below by the histograms, standardized as described above. Overlaid is the standard normal density. Across both regimes, the standardized histograms align closely with the Normal$(0, 1)$ curve, and the agreement improves as τ increases, providing a visual confirmation of the derived asymptotic distributions.

Simulation 2

To verify our interpretation of the data, we simulate daily temperatures

$$X_i = \mu_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{Normal}(0, \sigma^2),$$

with $\sigma = 20$, where the deterministic component follows

$$\mu_i = \begin{cases} \alpha, & i = 1, \ldots, 90 \quad \text{(January–February)}, \\ \alpha + \beta(i - 90), & i = 91, \ldots, 180 \quad \text{(March–April)}. \end{cases}$$

Bloom occurs when cumulative temperature first exceeds a threshold τ, i.e.

$$\nu(\tau) = \min\Big\{n \geq 1 : \sum_{i=1}^{n} X_i > \tau\Big\}.$$

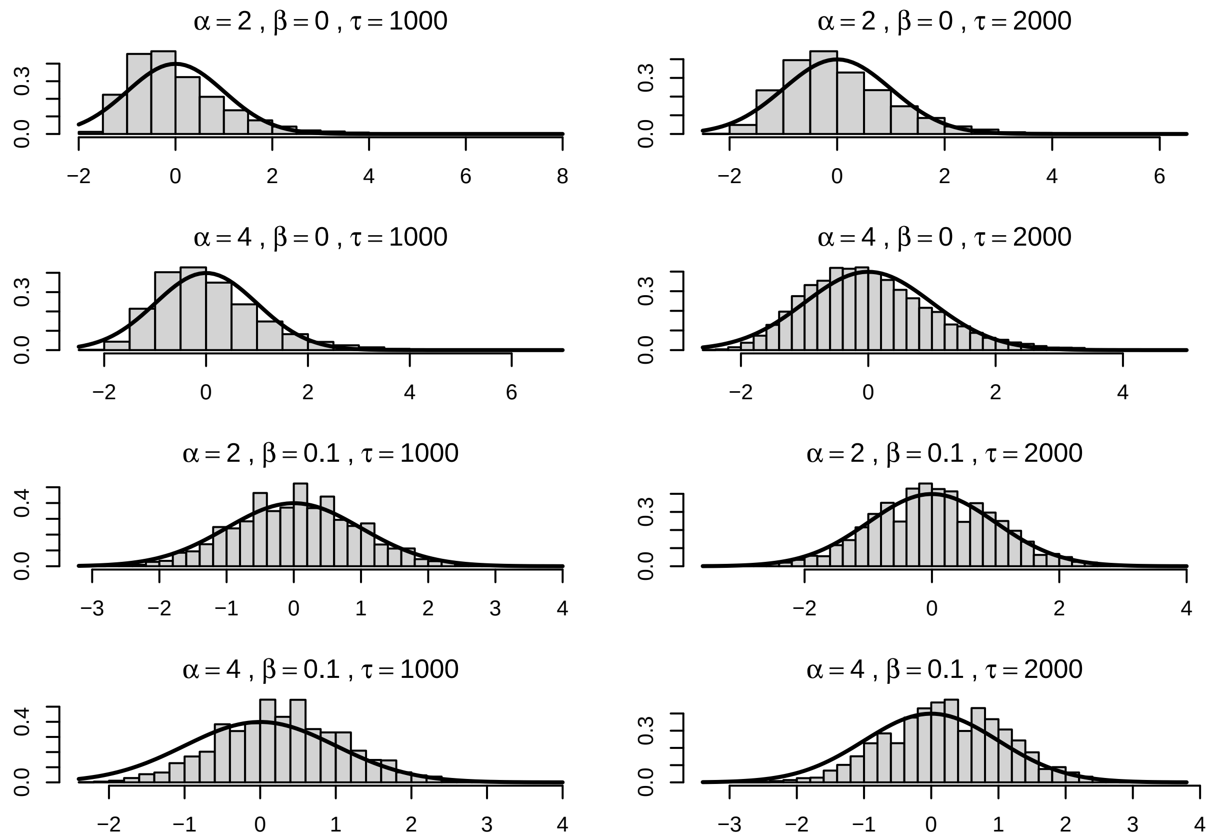

Figure 10: Simulations comparing the distribution of bloom dates (histogram) with an asymptotic approximation (solid line) under two regimes: winter temperatures (stationary temperatures, top) and spring warming (increasing temperatures, bottom).

Table 4: Simulated mean and standard deviation of the bloom date $\nu(t)$ across $R = 10{,}000$ replicates with thresholds $\tau \in \{1000, 2000\}$. Rows index the winter level α and columns index the spring warming rate β.

(a) mean, $\tau = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	151.66	136.83	124.86
8	114.87	111.27	106.85
10	98.45	97.20	95.72

(c) mean, $\tau = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	199.54	171.15	149.16
8	169.81	152.43	137.37
10	156.22	143.40	131.34

(b) sd, $\tau = 1000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	15.48	10.42	7.21
8	16.53	13.27	10.50
10	15.94	14.45	12.71

(d) sd, $\tau = 2000$

$\alpha \setminus \beta$	0.2	0.4	0.8
4	10.96	7.22	4.84
8	10.93	7.50	5.11
10	10.69	7.66	5.36

We simulated $R = 10{,}000$ independent realizations of $\nu(\tau)$ for each combination of $\alpha \in \{4, 8, 10\}$, $\beta \in \{0.2, 0.4, 0.8\}$, and $\tau \in \{1000, 2000\}$. We report the mean (top) and standard deviation (bottom) of $\nu(\tau)$ in Tables 4a-4d, arranged with rows indexing α and columns indexing β.

The results reproduce the qualitative patterns observed in the lilac data. Mean bloom dates decrease as either α increases or β increases (Tables 4a and 4c). Within a fixed α row, increasing β reduces the standard deviation of bloom dates (Tables 4b and 4d), consistent with the prediction that increasing average temperatures diminish the influence of noise on the threshold-crossing time. Holding β fixed, the standard deviation is often larger at higher α (particularly for $\beta \in \{0.4, 0.8\}$), consistent with the prediction that earlier crossings occur in a flatter portion of the seasonal cycle where microclimate variability has greater influence.